

Trees and clusters

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Tree-based methods

Decision trees can be used for **regression** and **classification** problems.

Example for regression: salary of baseball players as a function of years played and hits.

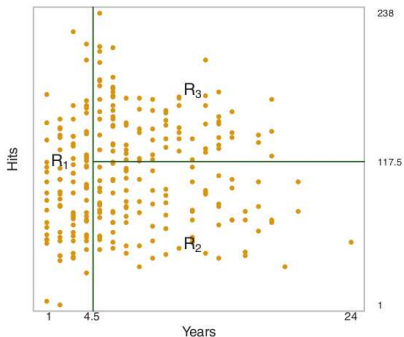
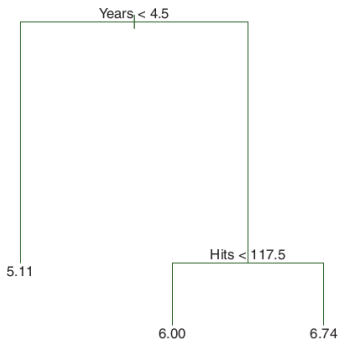


Figure from Jones et al. 2013

Regression trees

Prediction via stratification of the feature space

1. Divide the predictor space (set of possible values of $X_1, \dots, X_p$) into J distinct non-overlapping regions $R_1, R_2, \dots, R_J$.
2. For each observation in region R_j the prediction is the mean response values for the training observations in R_j .

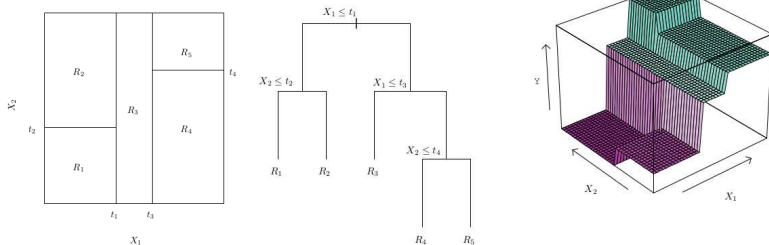


Figure from Jones et al. 2013

Stratifying the feature space

Aim: Find *boxes* that minimize the RSS (without searching through all possible partitions):

$$\sum_{j=1}^J \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

- Recursive binary splitting
 - ▶ Top-down - starts at the top of the tree
 - ▶ Greedy - at each step the best split is made at that particular step
 - ▶ Process stops when a criteria is reached (for example, no region contains more than h observations)

Step 1: select the predictor X_j and the cut point s such that the resulting tree has lowest RSS. Look for j and s to minimize:

$$\sum_{i: x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i: x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2$$

Repeat for each of the resulting regions.

Tree pruning

Aim: avoid overfitting and poor performance on test data by penalizing the number of terminal nodes $|T|$.

Algorithm:

1. Use recursive binary splitting to define a large tree on training data.
2. Apply cost complexity pruning to obtain a sequence of best subtrees for a given α
3. Use K-fold cross-validation to choose α
4. Return the tree from step 2, for the chosen α

Cost complexity pruning: For each α there is a subtree $T \subset T_0$ such that

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

is as small as possible.

Classification trees

- ▶ For each observation in region R_j the prediction is the most common occurring class in R_j .
- ▶ For tree growing regions are chosen to minimize a measure of node purity, **Gini index G** (or alternatively entropy).
- ▶ The prediction accuracy of the final pruned tree is estimated using the **classification error rate**

$$G = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk})$$

where $\hat{p}_{mk}$ is the proportion of training observations in the m region that are from the class k .

- ▶ G is small if all $\hat{p}_{mk}$ are close to 0 or 1, that is a node contains mainly observations from a single class.

Classification trees - example

Unpruned tree:

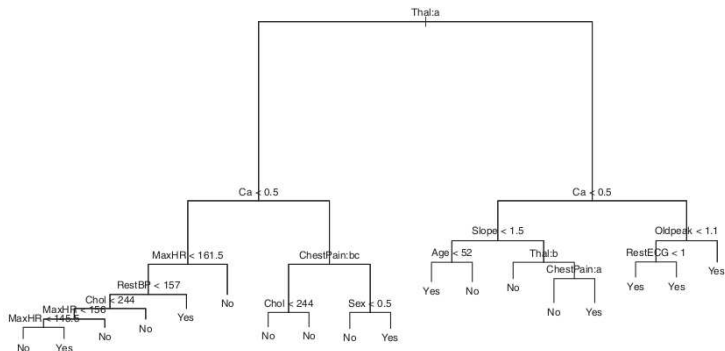
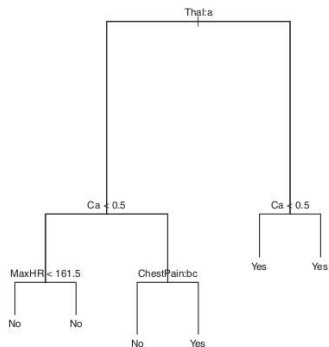
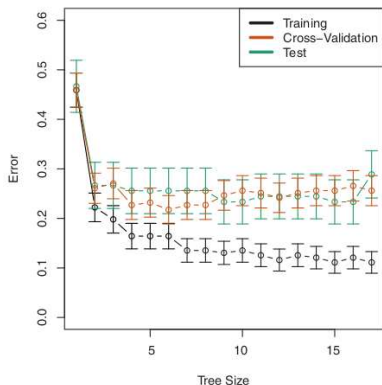


Figure from Jones et al. 2013

- ▶ outcome categorical variable: heart disease (Yes / No)
- ▶ 13 predictors, some categorical

Classification trees - example



- ▶ Some splits result in nodes with the same predicted value!
 - ▶ Those splits lead to increased node purity.
 - ▶ If a node is very pure, one will be more sure about the test prediction of an observation in that region.

Advantages and disadvantages of trees

- ▶ Easy and intuitive.
 - ▶ Easy to display graphically.
 - ▶ More like human decision-making.
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- ▶ Less good prediction accuracy than other methods.
 - ▶ Not very robust: a small change in data can result in a very different tree.

Bagging or bootstrap aggregation

Aim: Reduce the high variance of decision trees.

- ▶ Take B samples from the training data set (bootstrap).
 - ▶ B can be 100s or even 1000s
- ▶ Grow a tree model without pruning for each bootstrap.
 - ▶ Each tree has high variance but low bias.
- ▶ Average the B predictions, reducing the variance.
(Variance of the sample mean is σ^2/n .)

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*b}(x)$$

- ▶ The results are difficult to interpret - many different trees.
 - ▶ variable importance can be estimated
- ▶ Classification: use majority vote for prediction, predict the most commonly occurring class.

Out-of-bag error estimation

The test error can be estimated without cross-validation.

- ▶ The bootstraps use approximately $2/3$ of the observations and leave $1/3$ out observations out - out-of-bag observations.
- ▶ For each observation i the response can be predicted from all the bootstraps that did not include i .
- ▶ The OOB error is an estimate of the test error for the bagged model.
- ▶ Cross validation would be computationally expensive!

Random forests

Similar to bagging, but decorrelating the trees.

- ▶ Use B samples from the training data set (bootstrap).
- ▶ Grow a tree model for each bootstrap.
 - ▶ For each tree and each split, consider only m predictors chosen at random from the p possible predictors.
 - ▶ $m \approx \sqrt{p}$, (if $m = p$ bagging)
- ▶ The resulting trees will be more different from each other.
- ▶ The random forest predictions will be less variable and hence more reliable.

Unsupervised learning

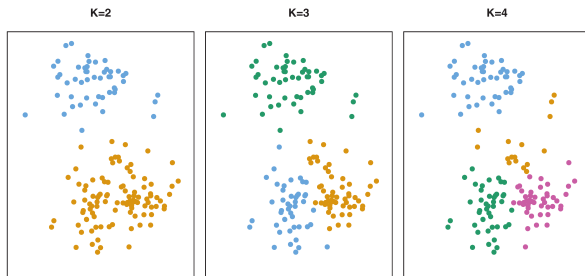
Unsupervised learning is a set of tools to explore possible structure in a set of n observation of p variables.

- ▶ There is no outcome variable - no predictions.
- ▶ There is no way to check how well the algorithm works.
- ▶ Often part of an exploratory data analysis.
 - ▶ Aiming at visualizing the data in informing ways - PCA.
 - ▶ Aiming at finding subgroups in the data - clustering.

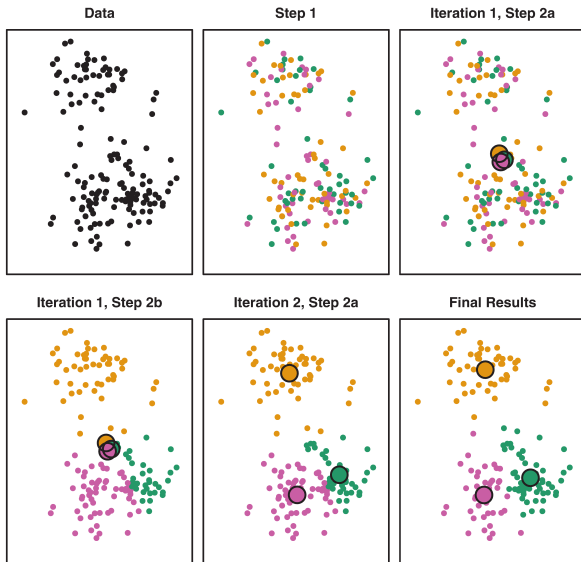
Clustering - K -means clustering

Aim: separating the observation into K clusters. c

- ▶ Each observation will be assigned to one and only one cluster - no overlapping.
- ▶ Clusters are determined so that the within cluster variation is as small as possible.
- ▶ Variation is defined using a distance - usually euclidean distance.
- ▶ The number K of clusters is pre-specified.



K-means clustering - one algorithm illustration



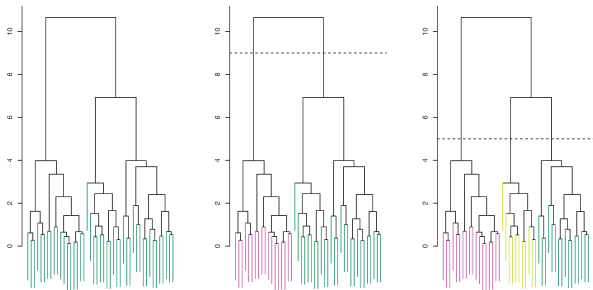
Local optimum

- ▶ The final solution is a local and not a global optimum.
- ▶ One runs several times and selects the best solution.



Hierarchical clustering

- ▶ Agglomerative clustering: clusters are fused, a pair at a time, based on similarity.
- ▶ The resulting dendrogram can be cut at different levels.
- ▶ A measure of distance is defined - euclidean distance, correlation
- ▶ A measure of cluster similarity is defined. For example: complete linkage or average linkage.



Hierarchical clustering

